

Student's Assessment Number _____

THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL OF TANZANIA
FORM TWO NATIONAL ASSESSMENT

031

PHYSICS

Time: 2:30 Hours

Year: 2025

Instructions

1. This paper consists of sections A, B and C with a total of **ten (10)** questions.
2. Answer **all** the questions.
3. All answers must be written in the spaces provided.
4. All writing must be in **blue** or **black** ink, **except** drawings which must be in pencil.
5. Communication devices and any unauthorised materials are **not** allowed in the assessment room.
6. Write your **Assessment Number** at the top right corner of every page.
7. Where necessary the following constants may be used:
 - (i) Acceleration due to gravity, $g = 10 \text{ m/s}^2$.
 - (ii) $\pi = 3.14$.

QUESTION NUMBER	FOR ASSESSOR'S USE ONLY	
	SCORE	ASSESSOR'S INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

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2

SECTION A (15 Marks)

Answer **all** questions in this section.

1. For each of the items (i) – (x), choose the correct answer among the given alternatives and write its letter in the box provided. **(10 marks)**

(i) Why is it important to study Physics in real life situations?

1. It is useful for career development
2. It is used to clarify the existence of historical sites
3. It explains physical properties of matter
4. It is applied in the manufacturing and designing of the electronic devices
5. It helps in experimental discoveries

A 1, 2, 3 and 4

B 2, 3, 4 and 5

C 1, 3, 4 and 5

D 1, 2, 4 and 5

- (ii) Figure 1 is a warning symbol used in the laboratory. What does the symbol represent?



Figure 1

- A The substance is corrosive
- B The substance catches fire easily
- C The substance oxidizes the air and speeds up burning
- D The substance produces radiations

- (iii) Which one is a renewable energy whose source is heat from the earth's interior?

A Wind energy

B Solar energy

C Geothermal energy

D Water energy

(iv) Imagine two groups of equal number of people are pulling a rope by their bare hands, each group is trying to overcome the opponent's force. What kind of force does each group exert on the rope?

- | | |
|--------------------|--------------------|
| A Tension force | B Normal force |
| C Frictional force | D Stretching force |

(v) What does a clinical thermometer measure?

- A Ice point of a solid
B Steam point of a liquid
C Temperature of a human body
D Room temperature

(vi) Why is hot soup tastier than cold soup?

- A It has higher surface tension
B It has lower surface tension
C It has higher capillary action
D It has lower capillary action

(vii) Which set represents ferromagnetic materials?

- A Aluminium, cobalt and nickel
B Chromium, cobalt and steel
C Steel, nickel and aluminium
D Cobalt, nickel and steel

(viii) A student was given a plane mirror and observed that its reflecting surface was so smooth that an image of the object that emits the light was very clear. Which type of reflection was a result of such observation?

- | | |
|----------------------|------------------------|
| A Diffuse reflection | B Reflection of light |
| C Regular reflection | D Selective absorption |

(ix) After coming late to school, a student ran into the classroom and closed the door to hide from the teacher. What type of force did the student use to close the door?

- | | |
|-----------------------|--------------------|
| A Normal force | B Turning force |
| C Compressional force | D Stretching force |

(x) If you want to detect the presence of electric charge in an object, which electrical device would you use?

- | | |
|-----------------|----------------|
| A Electrophorus | B Galvanometer |
| C Voltmeter | D Electroscope |

2. Match each of the physical quantities in **List A** with its corresponding units in **List B** by writing a letter of the correct answer below an item number in the table provided.

(5 marks)

List A	List B
(i) Temperature	A Ampere
(ii) Electric current	B Kilogram
(iii) Displacement	C Kelvin
(iv) Momentum	D Meter
(v) Mass	E Meter per second square
	F Kilogram meter per second
	G Mole

Answers

Item Number	(i)	(ii)	(iii)	(iv)	(v)
Answer					

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) Why a tractor with wide tyres cannot get stuck in muddy places as compared to a car with narrow tyres. **(2 marks)**

- (b) (i) List three factors on which the pressure of liquids depends. (3 marks)

- _____
- _____
- _____

- (ii) Calculate the area of a surface of an object which exerts a pressure of 0.2 N/m^2 when the force acting on it is 2 N . (5 marks)

4. (a) (i) When a body is totally immersed in a liquid, its weight becomes less than its weight in air. Why is the body weight decreases in a liquid?

(1 mark)

- (ii) If the density of object X was 1.4 g/cm^3 , predict if it will float in water. Give one reason to justify your choice. (4 marks)

- (b) The weight of a piece of aluminium in air is 42.0 N and 25.5 N when completely immersed in water. What is the relative density of aluminium? (5 marks)

5. (a) State the Newton's third law of motion. (3 marks)

Student's Assessment Number _____

- (b) A monkey has a mass of 45 kg, and it climbs on a rope which can stand a maximum tensional force of 550 N. If the monkey climbs up with an acceleration of 5.5 m/s^2 , find the force that will make the rope to break?

(7 marks)

6. (a) Write three equations of uniformly accelerated motion.

(3 marks)

(i) _____

(ii) _____

Student's Assessment Number _____

(iii)

(b) A car started to move from rest and accelerated uniformly at the rate of 4 m/s^2 for 12 s. It then maintained a constant speed for 30 seconds, after applying the brakes, it decelerated uniformly to rest in 5 s. Calculate;

(i) The maximum speed reached. (2 marks)

(i) The maximum speed reached. (2 marks)

(ii) The total distance covered in metres. (5 marks)

(ii) The total distance covered in metres. (5 marks)

Student's Assessment Number _____

7. (a) With well labelled diagrams, differentiate a fixed from a movable pulley. Give two differences. **(5 marks)**

- (b) A pulley system has mechanical advantage of 4. An effort of 100 N is applied on the pulleys system. If the pulley system has an efficiency of 75%, determine the maximum load that will be raised by the effort applied. **(5 marks)**

8. (a) Outline five forms of energy. (2.5 marks)

- (i) _____
- (ii) _____
- (iii) _____
- (iv) _____
- (v) _____

(b) An object of mass 200 kg is lifted by a crane to the top of a building that is 12 m high in 10 s. Calculate the power of the crane. (7.5 marks)

9. (a) State systematically six steps involved in measuring the volume of an irregular object by using the Eureka can. (3 marks)

1. General Information
 2. Project Description
 3. Objectives
 4. Methodology
 5. Results
 6. Conclusion
 7. References
 8. Appendix
 9. Index
 10. Table of Contents
 11. Abstract
 12. Introduction
 13. Background
 14. Statement of the Problem
 15. Significance of the Study
 16. Scope of the Study
 17. Limitations of the Study
 18. Organization of the Study
 19. Summary
 20. References
 21. Appendix
 22. Index
 23. Table of Contents
 24. Abstract
 25. Introduction
 26. Background
 27. Statement of the Problem
 28. Significance of the Study
 29. Scope of the Study
 30. Limitations of the Study
 31. Organization of the Study
 32. Summary
 33. References
 34. Appendix
 35. Index
 36. Table of Contents
 37. Abstract
 38. Introduction
 39. Background
 40. Statement of the Problem
 41. Significance of the Study
 42. Scope of the Study
 43. Limitations of the Study
 44. Organization of the Study
 45. Summary
 46. References
 47. Appendix
 48. Index
 49. Table of Contents
 50. Abstract
 51. Introduction
 52. Background
 53. Statement of the Problem
 54. Significance of the Study
 55. Scope of the Study
 56. Limitations of the Study
 57. Organization of the Study
 58. Summary
 59. References
 60. Appendix
 61. Index
 62. Table of Contents
 63. Abstract
 64. Introduction
 65. Background
 66. Statement of the Problem
 67. Significance of the Study
 68. Scope of the Study
 69. Limitations of the Study
 70. Organization of the Study
 71. Summary
 72. References
 73. Appendix
 74. Index
 75. Table of Contents
 76. Abstract
 77. Introduction
 78. Background
 79. Statement of the Problem
 80. Significance of the Study
 81. Scope of the Study
 82. Limitations of the Study
 83. Organization of the Study
 84. Summary
 85. References
 86. Appendix
 87. Index
 88. Table of Contents
 89. Abstract
 90. Introduction
 91. Background
 92. Statement of the Problem
 93. Significance of the Study
 94. Scope of the Study
 95. Limitations of the Study
 96. Organization of the Study
 97. Summary
 98. References
 99. Appendix
 100. Index
 101. Table of Contents
 102. Abstract
 103. Introduction
 104. Background
 105. Statement of the Problem
 106. Significance of the Study
 107. Scope of the Study
 108. Limitations of the Study
 109. Organization of the Study
 110. Summary
 111. References
 112. Appendix
 113. Index
 114. Table of Contents
 115. Abstract
 116. Introduction
 117. Background
 118. Statement of the Problem
 119. Significance of the Study
 120. Scope of the Study
 121. Limitations of the Study
 122. Organization of the Study
 123. Summary
 124. References
 125. Appendix
 126. Index
 127. Table of Contents
 128. Abstract
 129. Introduction
 130. Background
 131. Statement of the Problem
 132. Significance of the Study
 133. Scope of the Study
 134. Limitations of the Study
 135. Organization of the Study
 136. Summary
 137. References
 138. Appendix
 139. Index
 140. Table of Contents
 141. Abstract
 142. Introduction
 143. Background
 144. Statement of the Problem
 145. Significance of the Study
 146. Scope of the Study
 147. Limitations of the Study
 148. Organization of the Study
 149. Summary
 150. References
 151. Appendix
 152. Index
 153. Table of Contents
 154. Abstract
 155. Introduction
 156. Background
 157. Statement of the Problem
 158. Significance of the Study
 159. Scope of the Study
 160. Limitations of the Study
 161. Organization of the Study
 162. Summary
 163. References
 164. Appendix
 165. Index
 166. Table of Contents
 167. Abstract
 168. Introduction
 169. Background
 170. Statement of the Problem
 171. Significance of the Study
 172. Scope of the Study
 173. Limitations of the Study
 174. Organization of the Study
 175. Summary
 176. References
 177. Appendix
 178. Index
 179. Table of Contents
 180. Abstract
 181. Introduction
 182. Background
 183. Statement of the Problem
 184. Significance of the Study
 185. Scope of the Study
 186. Limitations of the Study
 187. Organization of the Study
 188. Summary
 189. References
 190. Appendix
 191. Index
 192. Table of Contents
 193. Abstract
 194. Introduction
 195. Background
 196. Statement of the Problem
 197. Significance of the Study
 198. Scope of the Study
 199. Limitations of the Study
 200. Organization of the Study
 201. Summary
 202. References
 203. Appendix
 204. Index
 205. Table of Contents
 206. Abstract
 207. Introduction
 208. Background
 209. Statement of the Problem
 210. Significance of the Study
 211. Scope of the Study
 212. Limitations of the Study
 213. Organization of the Study
 214. Summary
 215. References
 216. Appendix
 217. Index
 218. Table of Contents
 219. Abstract
 220. Introduction
 221. Background
 222. Statement of the Problem
 223. Significance of the Study
 224. Scope of the Study
 225. Limitations of the Study
 226. Organization of the Study
 227. Summary
 228. References
 229. Appendix
 230. Index
 231. Table of Contents
 232. Abstract
 233. Introduction
 234. Background
 235. Statement of the Problem
 236. Significance of the Study
 237. Scope of the Study
 238. Limitations of the Study
 239. Organization of the Study
 240. Summary<

- (b) (i) What is the function of the micrometer screw gauge? (2 marks)

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- (ii) Draw a micrometer screw gauge and label its four parts. (5 marks)

SECTION C (15 Marks)

Answer question ten (10).

10. (a) (i) State Ohm's law. (1 mark)

- (ii) Outline the procedures of conducting an experiment to verify Ohm's law.
Give at least five necessary procedures. (5 marks)

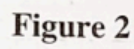
- _____
- _____
- _____
- _____
- _____

- (b) Show that the total resistance of two resistors connected in parallel is given by

$R_T = \frac{R_1 R_2}{R_1 + R_2}$ where R_1 and R_2 are the resistances of individual resistors.

(4 marks)

(c) Consider the electric circuit in Figure 2.



- (3 marks)

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